Economic, Environmental, and Equity Outcomes Assessment of the Electric Vehicle Charging Network Expansion in Illinois
R27-280
Quarterly Report

ILLINOIS

ILLINOIS CENTER FOR TRANSPORTATION

Progress Status (Tasks Completed / Ongoing)

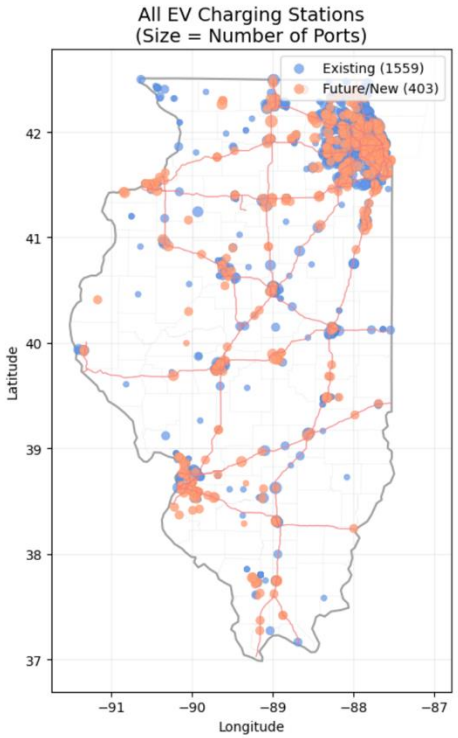
Task 1.1 – Data Acquisition and Integration ✓ Completed

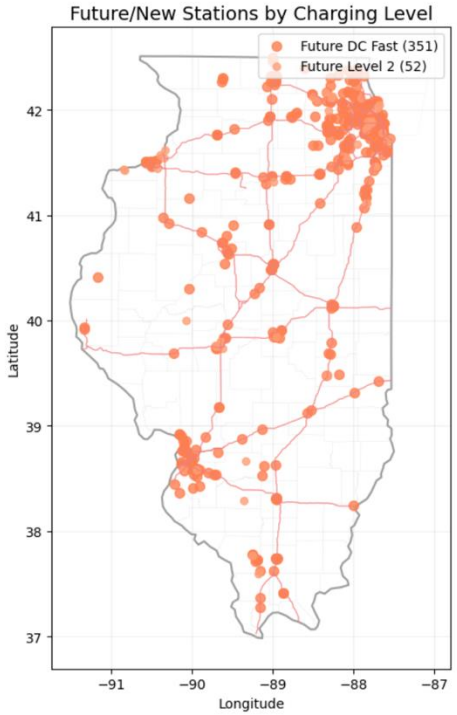
Task 1.2 – Gap and Accessibility Analysis ✓ Completed (inaccessibility framework with spillover capacity)

Task 1.3 – Electrified Miles Estimation ▶ In progress (preliminary model and code developed)

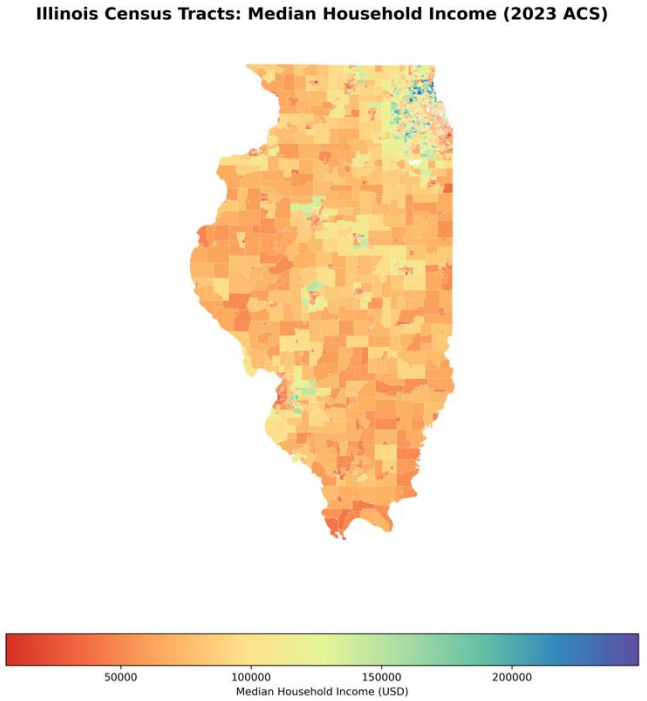
Tasks 1.4 and Objective 2+ scheduled for upcoming quarters

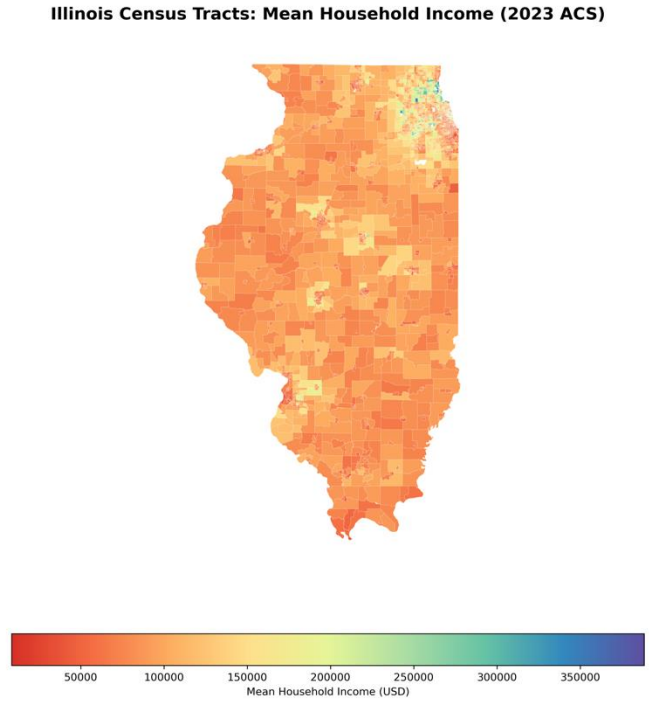
Data & Visualizations Developed





Current and planned EV charging stations





Tract-level household income distribution

Task 1.2 Gap and Accessibility Analysis

For each census tract j, the number of EVs that can be served, T_j, is calculated based on charger availability, charging rate, utilization, and average daily driving demand. Specifically, the service potential of each Level 2 and DCFC charger is estimated as:

$$E2_{i_2,j} = \frac{C_2 U_2}{D_j A_{i_2,k}}, \quad E3_{i_3,j} = \frac{C_3 U_3}{D_j A_{i_3,k}},$$

where:

- C_2, C_3 = charging rates of Level 2 and DCFC ports,
- U_2, U_3 = average daily utilization hours of Level 2 and DCFC ports,
- D_j = average daily vehicle miles traveled per EV in census tract j,
- $A_{i_2,k}, A_{i_3,k}$ = number of vehicles in the service area of station k with Level 2 or DCFC charging.

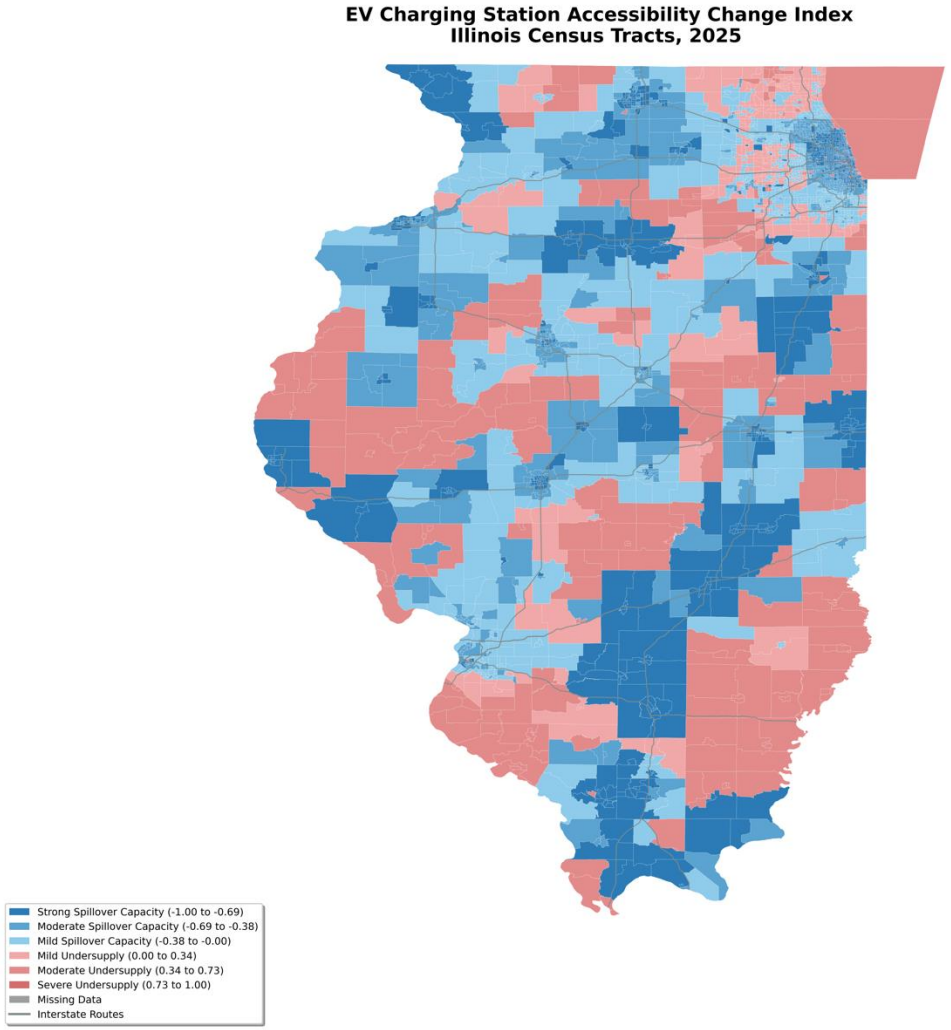
$$IA_j = \frac{V_j - T_j}{V_j + T_j},$$

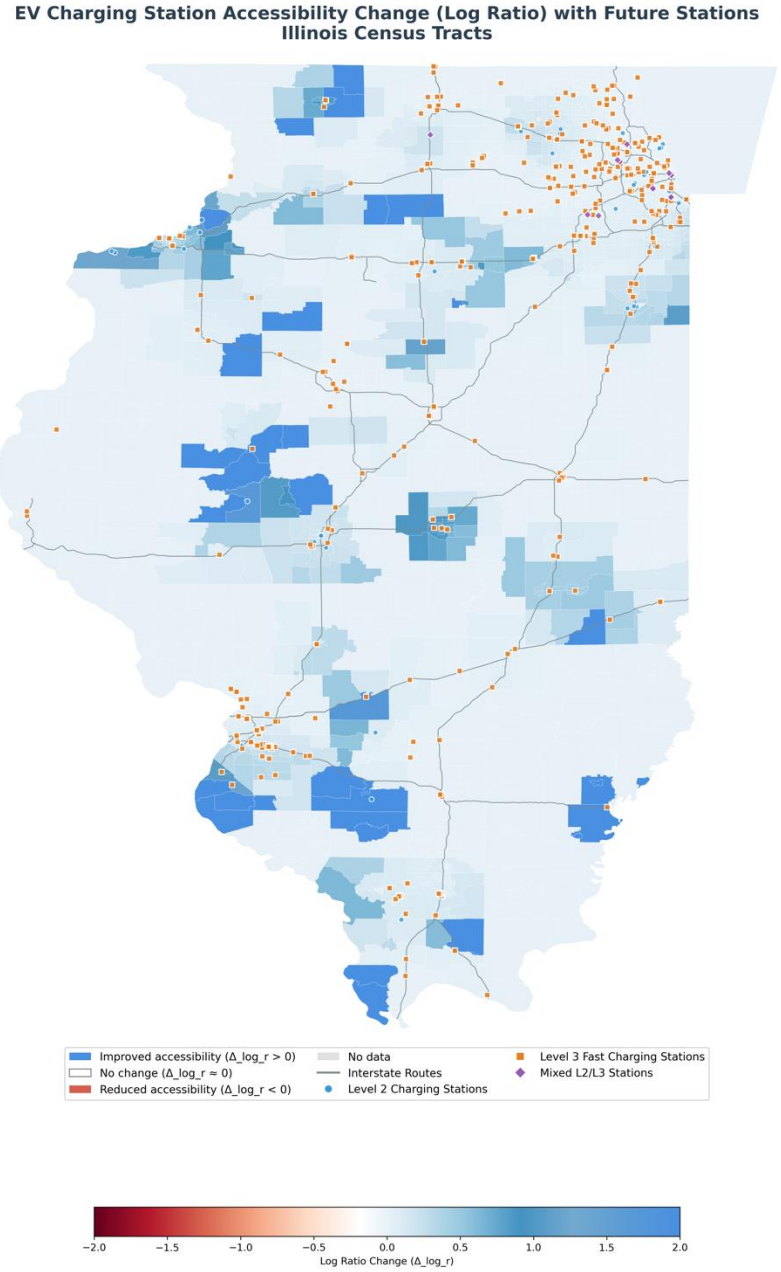
The tract-level serviceable EVs are then aggregated as:

$$T_j = Pop_j \left(\sum_{i_2=1}^{N_{2,j}} E2_{i_2,j} + \sum_{i_3=1}^{N_{3,j}} E3_{i_3,j} \right),$$

where $N_{2,j}$ and $N_{3,j}$ are the numbers of Level 2 and DCFC ports in tract j, and Pop_j is the tract population scaling factor.

Task 1.2 Gap and Accessibility Analysis





Anticipated Work Next Quarter

- Task 1.3 – Electrified Miles Estimation
  - Complete modeling of additional electrified VMT from station expansion
  - Incorporate heterogeneous travel patterns, with rural and underserved focus
  - Produce tract- and county-level electrified miles outputs

- Task 1.4 – Worth of EV Charging
  - Translate electrified VMT gains into driver benefits (fuel cost, detour, time savings)
  - Use EIA fuel price data and EPA/DOE performance metrics for calculations

- Objective 2 Preparatory Work
  - Design econometric framework (linking charging investments to jobs, GRP, business activity)
  - Collect and integrate data sources (BEA, BLS, AFLEET)
  - Draft structure for future stakeholder engagement (Task 2.2 Virtual Townhall)


ILLINOIS